

TROUBLE SHOOTING AND HELPDESK

TASK: 1

A user reports their laptop won't power on even when plugged in. List 5 troubleshooting steps you would take to diagnose the cause, and for each step, explain what you are checking for.

Laptop Won't Power On – Troubleshooting Steps

1. **Check the Power Adapter and Wall Socket:**
First, check whether the charger is properly connected to the laptop and the wall socket. Try a different wall socket and check for any damage to the charger cable. This checks whether the problem is caused by a faulty power supply or charger.
2. **Check the Charging Indicator:**
Connect the charger and observe the laptop's charging LED. If there is no light, it may indicate a problem with the charger, charging port, or power circuit. If the light is present, the laptop is receiving some power.
3. **Perform a Hard Reset:**
Disconnect the charger and remove the battery if it is removable. Hold the power button for about 15–30 seconds, then reconnect the charger and try powering on the laptop. This checks whether a temporary power or hardware state is preventing startup.
4. **Check the Battery:**
If possible, test the laptop using only the AC adapter or with a known-good battery. This checks whether the battery is completely discharged, damaged, or preventing the laptop from receiving power.
5. **Inspect Internal Hardware and Seek Service:**
If the laptop still does not power on, check for loose RAM, damaged connectors, or signs of motherboard damage, if safe to do so. If there are no obvious problems, professional service may be required to test the motherboard and power circuit.

TASK: 2

You receive a helpdesk ticket: 'My monitor is showing a blank screen, but the CPU seems to be running.' Write a step-by-step troubleshooting flow you would follow to identify and resolve the issue.

Hint: Consider checking both hardware connections and basic display settings.

Monitor Blank Screen – Troubleshooting Flow

- 1. Check Monitor Power:**
Make sure the monitor is switched on and the power cable is properly connected. Check whether the monitor power LED is glowing.
- 2. Check Display Cable:**
Check the HDMI, DisplayPort, VGA, or DVI cable connecting the computer to the monitor. Ensure both ends are firmly connected. If possible, try another cable.
- 3. Check the Correct Input Source:**
Open the monitor's input/source menu and select the port being used, such as HDMI or DisplayPort. An incorrect input source can cause a blank screen.
- 4. Check the Computer's Display Connection:**
If the computer has a dedicated graphics card, make sure the monitor cable is connected to the graphics card's video output rather than the motherboard's port.
- 5. Test With Another Monitor:**
Connect the computer to another working monitor or TV. If the second display works, the original monitor or its cable may be faulty.
- 6. Check Display Settings:**
If the computer starts but the display is not visible, use the Windows shortcut **Windows + P** and select **PC screen only** or **Duplicate**. This checks whether the display is set to an incorrect output mode.
- 7. Check Graphics Driver:**
If Windows loads but the screen remains blank, restart the computer and check the graphics driver using Device Manager. Update or reinstall the graphics driver if necessary.

8. Restart and Confirm the Fix:

Restart the computer after making the changes. Confirm that the monitor displays the normal startup screen and Windows desktop.

Conclusion:

The problem is most commonly caused by a loose/incorrect display connection, wrong monitor input, incorrect display settings, faulty cable/monitor, or a graphics-card/driver issue. The faulty component can be identified by testing each part systematically.

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TASK: 3

A user complains that their wireless mouse is unresponsive, but the keyboard works fine. Describe how you would isolate whether the issue is with the mouse, the USB port, or the system settings.

Wireless Mouse Troubleshooting

1. **Check the Mouse Power and Battery:**
First, check whether the wireless mouse is switched on and the battery is properly inserted. Replace the battery if it is weak or dead. This checks whether the mouse itself has a power problem.
2. **Check the USB Receiver:**
Make sure the wireless mouse USB receiver is firmly connected to the computer. Remove it and reconnect it to ensure a proper connection.
3. **Test Another USB Port:**
Connect the mouse receiver to a different USB port. If the mouse works on another port, the original USB port may be faulty or disabled.
4. **Test the Mouse on Another Computer:**
Connect the mouse receiver to another computer. If the mouse still does not work, the mouse or receiver is likely faulty. If it works, the problem is probably with the original computer.
5. **Check System Settings and Drivers:**
Open **Device Manager** → **Mice and other pointing devices** and check whether the mouse is detected or has a warning symbol. Update or reinstall the mouse/HID driver if required. Also check Windows Bluetooth settings if the mouse uses Bluetooth instead of a USB receiver.
6. **Restart the Computer:**
Restart the system and test the mouse again. This can resolve temporary USB, driver, or system-setting problems.

Conclusion:

By testing the mouse on another USB port and another computer, the problem can be isolated. If it fails everywhere, the mouse or receiver is likely faulty. If it works elsewhere, the original USB port, driver, or system settings are likely responsible.

TASK: 4

Imagine you are working on a helpdesk team for a gaming cafe. One system is overheating and shutting down during use. List 3 possible hardware causes and describe how you would check each one.

Hint: Think about fans, dust, and thermal paste.

Gaming PC Overheating – Hardware Causes and Checks

1. **Faulty or Non-Working Fans:**

Check whether the CPU, GPU, and cabinet fans are spinning properly while the system is running. A stopped or slow fan can cause heat to build up. Replace the faulty fan if necessary.

2. **Dust Buildup:**

Open the PC cabinet after switching it off and check the fans, air vents, heatsinks, and filters for dust. Excessive dust can block airflow and prevent proper cooling. Clean the components carefully using compressed air.

3. **Dried or Improper Thermal Paste:**

Check the thermal paste between the CPU and its heatsink. Old, dried, or insufficient thermal paste can reduce heat transfer from the CPU to the heatsink. Clean the old paste and apply a suitable new layer before reinstalling the cooler.

Conclusion:

I would check the fans first, then clean any dust buildup, and finally inspect and replace the thermal paste. After each repair, I would monitor the CPU/GPU temperature during gaming to confirm that the overheating and shutdown problem is resolved.